Jiayi Guo

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Education

University of California, San Diego San Diego, CA
M.S. in Computer Science *Sep. 2026 – Jun. 2028*

University of Electronic Science and Technology of China & University of Glasgow Chengdu, China
B.Eng. in Communication Engineering, GPA: 3.72/4.00 *Sep. 2022 – Jul. 2026*

Technical Skills

Languages: Python, C, MATLAB **ML/Data:** PyTorch, NumPy, SciPy, Pandas, Matplotlib, tabular ML, model evaluation
Engineering: FastAPI, Docker, REST APIs, JSON, Git, LaTeX **AI Areas:** LLM evaluation, RAG, diffusion models, AutoML

Publications

- **J. Guo**, L. Zhang, and Y. Shen, "A Human-Centered Multimodal AutoML Framework for Discriminative and Generative Tasks with Large Language Models," IEEE BIBM 2025.
- L. Zhang and **J. Guo**, "When Do LLM Agents Treat Surface Noise Differently from Semantic Noise? A 68-Cell Measurement Study with a Held-Out Trace-Level Validation," submitted to EMNLP 2026.

Projects

Resource-Aware LLM/RAG Evaluation Service *Python, FastAPI, Docker*

– Built a local-first LLM/RAG evaluation service with lightweight document retrieval, JSON benchmark schema, FastAPI endpoint, and Markdown/HTML reports.
– Designed metrics for answer quality, context relevance, hallucination-risk proxy, latency, and failure categories; added optional DeepSeek LLM judge mode.

Kaggle Playground Series: Predicting Irrigation Need *Python, Pandas, PyTorch*

– Built a tabular ML pipeline for irrigation-need prediction, covering preprocessing, feature engineering, model training, validation, and submission optimization.
– Ranked 16th out of 4,315 participants, placing in the top 3% of the competition.

Advertiser Growth CRM Backend Extension *Java, Spring Boot, PostgreSQL, Redis, Kafka, Docker*

– Extended the Apache-2.0 Spring PetClinic REST baseline with advertiser-account CRUD, paginated opportunity search, RBAC, optimistic locking, and a guarded sales-stage lifecycle.
– Implemented idempotent stage transitions and a transactional outbox that atomically persists business state and audit events, with bounded Kafka retry and observable failure states.
– Added Redis-backed account caching with update invalidation and database fallback, plus PostgreSQL persistence, validation, unified error responses, Actuator health checks, and Prometheus metrics.

Trustworthy Code-Change Review Agent with Evaluation and Observability

– Extended Microsoft Agent Framework Samples into a typed multi-agent code-review service that decomposes requirements, risk analysis, test planning, and report synthesis across concurrent Python workflows.
– Built FastAPI and CLI interfaces with 100 KiB input limits, atomic JSON persistence, structured failure states, health checks, and operational metrics; verified a complete create/fetch/trace HTTP path locally.
– Implemented Responsible AI controls that redact credentials and PII before model calls and block prompt-injection content, while keeping raw diffs and prompts out of OpenTelemetry attributes.
– Designed 27 unit/integration tests covering validation, redaction, DeepSeek/provider configuration and provenance, evaluation preflight, provider failures, configurable timeouts, malformed agent output, persistence, API contracts, and current Microsoft Agent Framework API compatibility.

Research Experience

Large Language Models for Automated Machine Learning *Johns Hopkins University, Feb. 2024 – May 2025*

– Led an independent research project applying LLMs to automate discriminative and generative ML workflows across multiple datasets and task settings.
– Evaluated framework performance under different conditions and conducted a 25-participant user study; wrote the paper accepted by IEEE BIBM 2025.

Diffusion Models for Virtual Try-On Systems *National Supercomputing Center, Sep. 2024 – Apr. 2025*

– Developed a diffusion-based virtual try-on framework for single-image garment replacement, including staged and partial replacement for sleeve changes.
– Improved visual realism and garment fitting quality for online-shopping try-on scenarios through model and pipeline adjustments.

Generative AI in Semantic Communication Systems *UESTC, Apr. 2024 – May 2025*

– Explored generative AI methods for latent-space compression and reconstruction in communication systems, including underwater signal recovery settings.
– Adapted AI models to complex-valued communication scenarios where standard real-valued modeling assumptions do not directly apply.